Cellular Adaptation & Reversible Injury

MDP 10208

Learning objectives

At the end of the teaching session, the student should be able to:

- Describe differences between hyperplasia, hypertrophy, atrophy and metaplasia.
- Explain causes and mechanism of cell injury.
- Describe reversible cell injury like: fatty change, hydropic degeneration and briefly about intracellular accumulations.
- Describe different types of intracellular cellular accumulations.

CASE 1

- **A 15-year-old boy fractured the bones of his right lower leg while playing football.**
  His leg was immobilised in a plaster cast.
- **After 6 weeks**, when the plaster was removed, he noticed that his **right leg looked thinner** than the left.
  The doctor advised rehabilitation exercises.





Q1. Define the pathological term which can describe the condition of his left leg.

Q2. State the cause for thinning of his left lower leg.

Q3. Describe briefly the pathological mechanism involved in this condition.

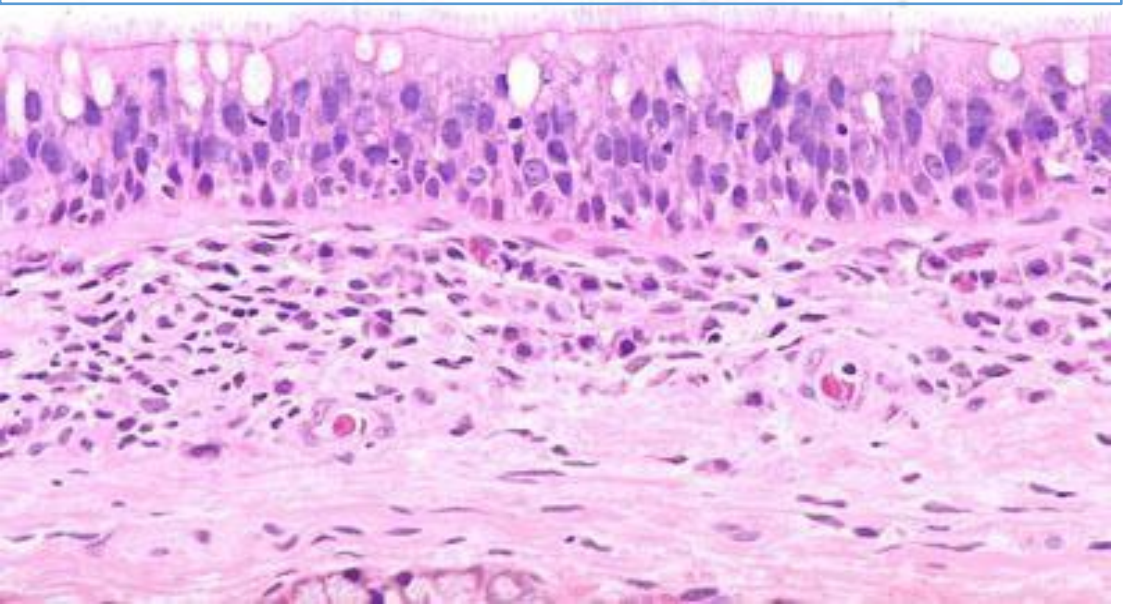
Q4. List FOUR other causes that can lead to this type of pathology.

CASE 2

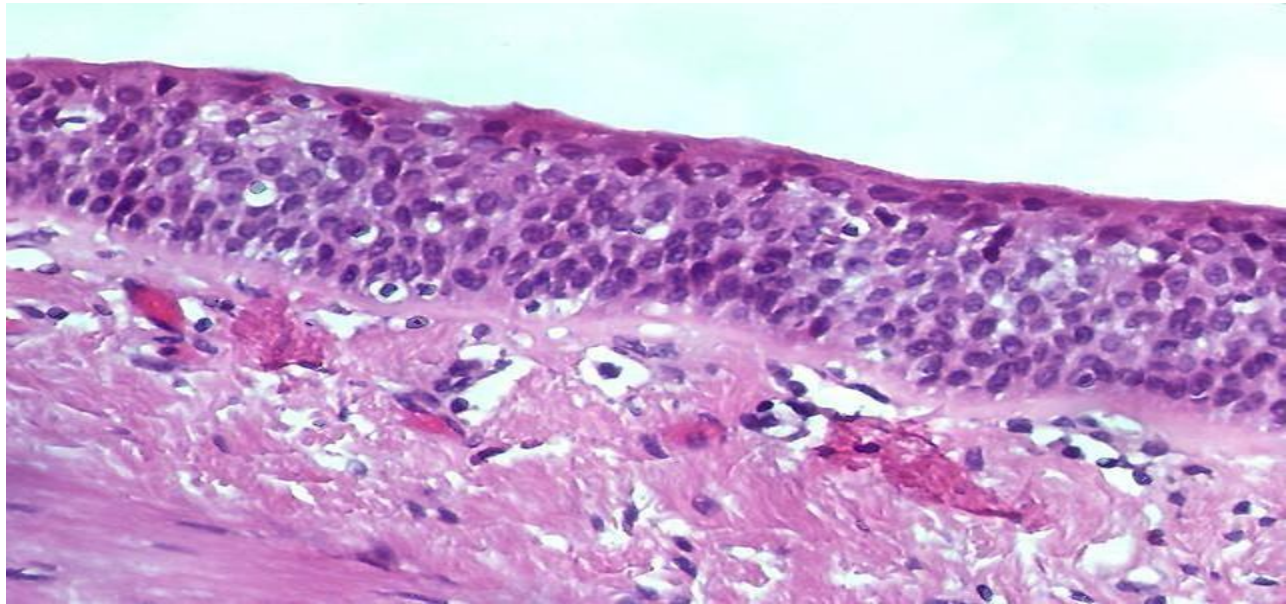
A 56-year-old man with twenty years history of smoking complaint of haemoptysis (coughing up of blood).

Bronchial biopsy was performed. Bronchial biopsy showed normal pseudostratified ciliated columnar epithelium was replaced with mature stratified squamous epithelium.

Normal bronchial lining epithelium



Patient's bronchial lining epithelium



Q1. Define the pathological term which can describe the condition of his bronchial tissue.

Q2. Describe the underlying mechanism for your answer in Q 1

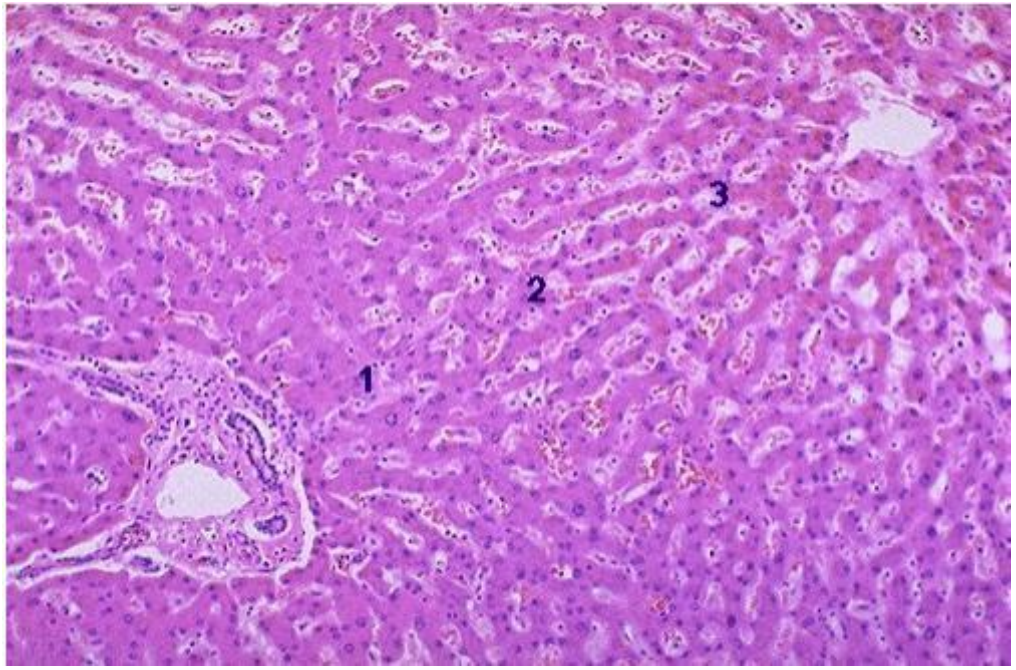
Q3. Describe THREE other examples of tissue that undergo this type of pathological changes.

Q4. Discuss TWO other types of cellular adaptation to environmental stress. Give on example for each type.

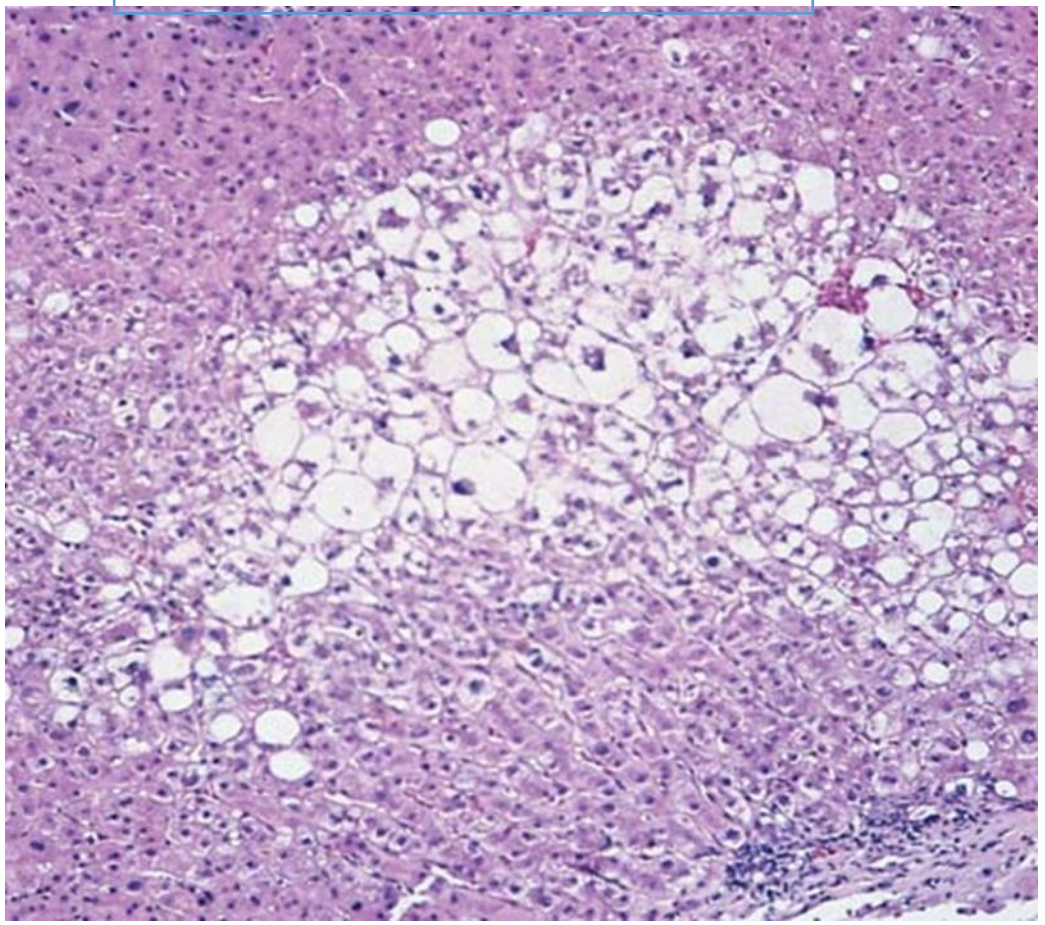
CASE 3

A 38-year-old man was recently started on high-dose paracetamol for severe back pain. After 3 days, he developed nausea and right upper quadrant discomfort. Liver biopsy was performed

Normal liver tissue



Patient's liver tissue



Q1. Explain the cellular changes in patients' liver disease.

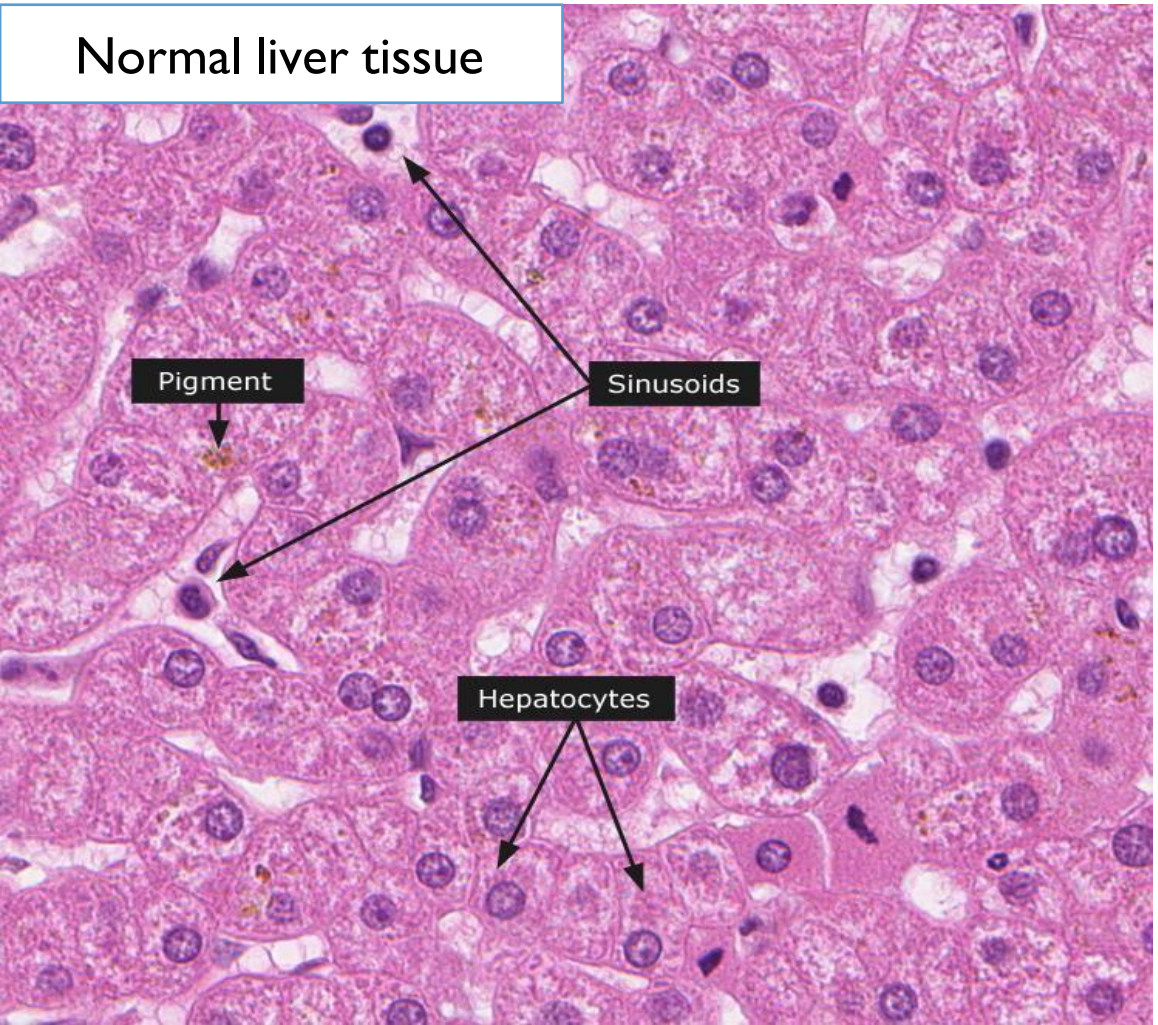
Q2. State the type of cell injury seen in this case.

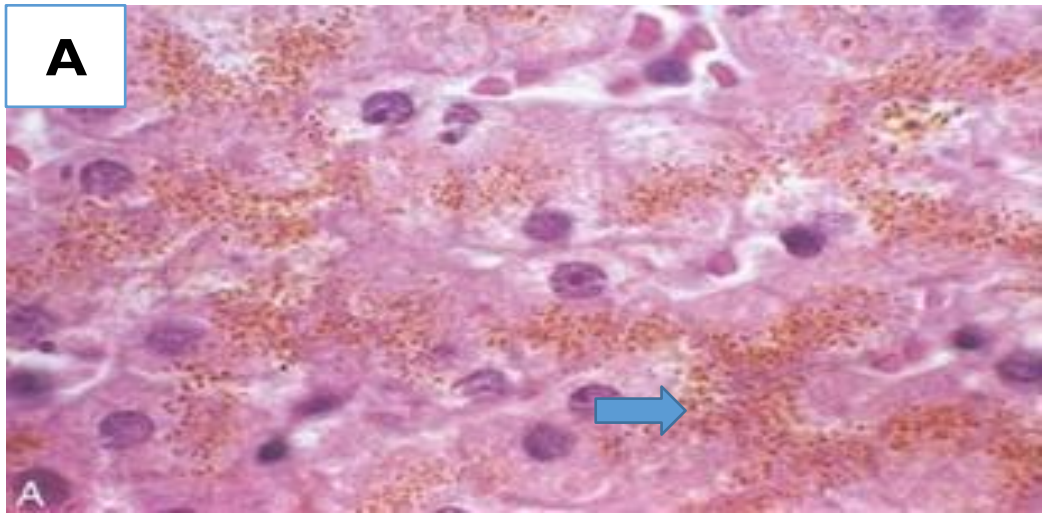
Q3. With one example, discuss the causes of cell injury.

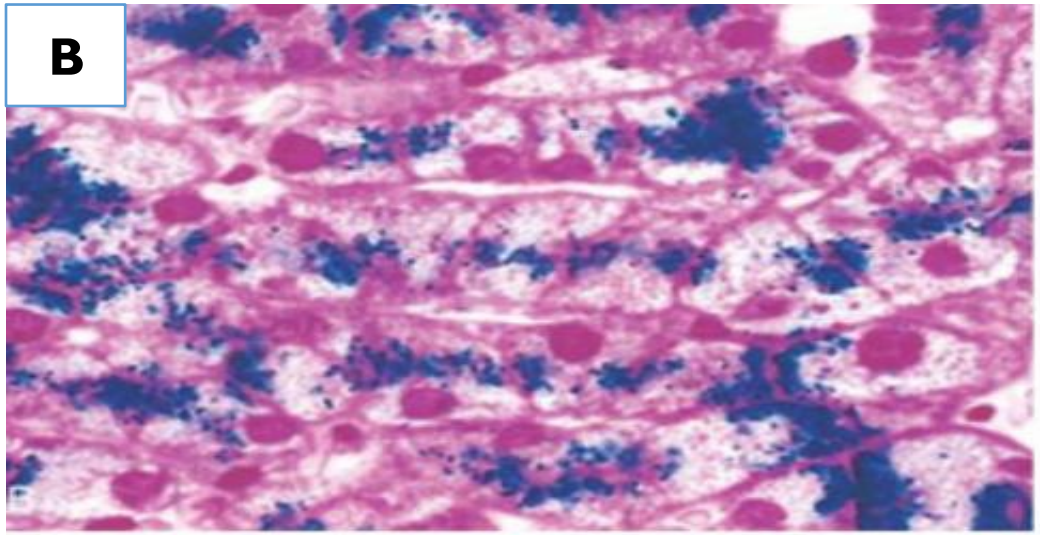
Q4. List the biochemical mechanisms of cell injury.

CASE 4

A 12-year-old boy, a known case of thalassemia major, with a history of receiving multiple blood transfusion presented with increased abdominal girth, yellow discolouration of his sclera and skin and liver enlargement. The haematoxylin and eosin (H & E) stain (A) and Prussian blue stain (B) of his liver biopsy are shown in the image.







Q1. State the type of intracellular pigment seen in this case.

Q2.Explain the underlying mechanism of his condition.

Q3. With one example, describe other types of intracellular pigments.

Case scenario

(Irreversible cell injury and tissue degeneration)

Learning objectives

At the end of the teaching session, the student should be able to:

- Describe irreversible cell injury.
- List the different types of necrosis and their morphological changes.
- Describe cell death due to apoptosis and its differences from necrosis.
- List differences between dystrophic and metastatic calcification.

CASE 1



Case Summary

59-year-old HIV-positive man

Presented with high fever

Developed acute bronchopneumonia

Died of respiratory insufficiency

Autopsy Finding

→ **Lung abscess** (yellow circle)

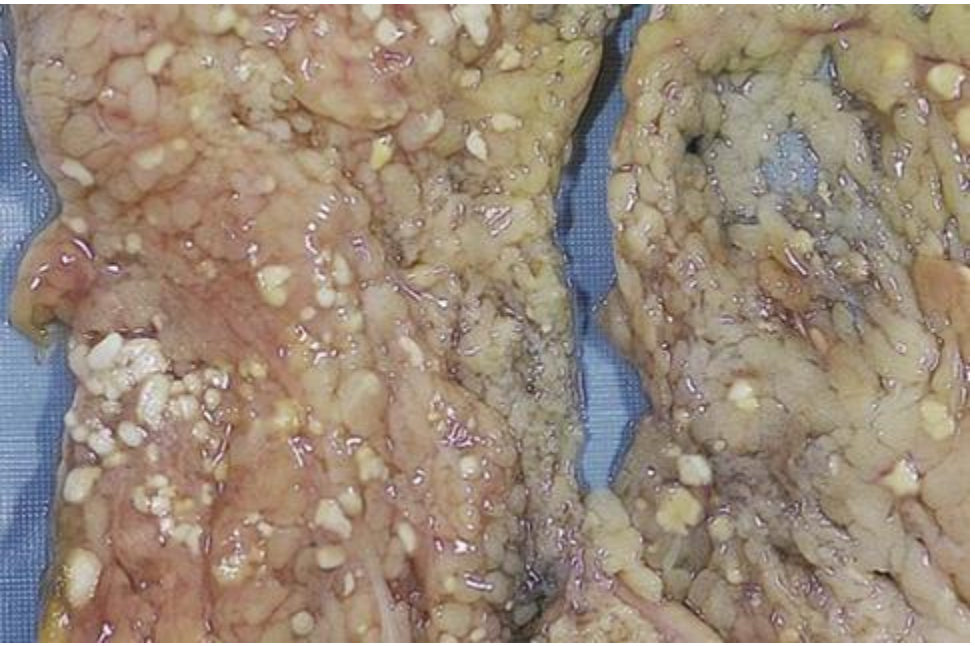
Q1. State the type of necrosis seen in this case.

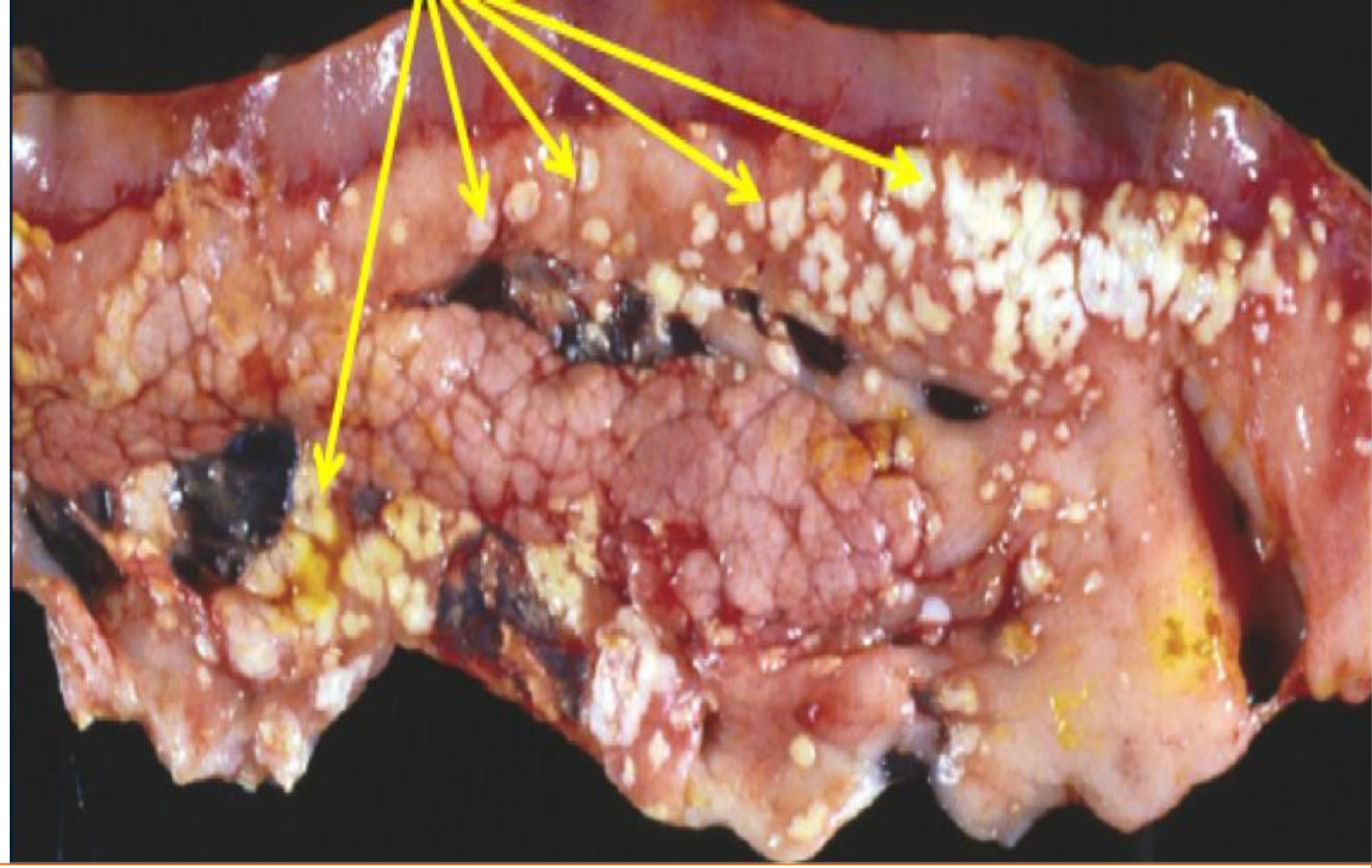
Q2. Describe briefly the THREE nuclear patterns of necrotic cells.

Q3. Describe the underlying mechanism of these pathological change.

Q4. State ONE OTHER example of similar pathology.

CASE 2





A pancreas from a 59-year-old man with long-standing alcoholism who presented with acute pancreatitis. Multiple chalky white deposits of fat necrosis are seen.

Q1. State the type of necrosis seen in this case.

Q2. Describe the underlying mechanism of these pathological change.

Q3. State ONE OTHER example of similar pathology.

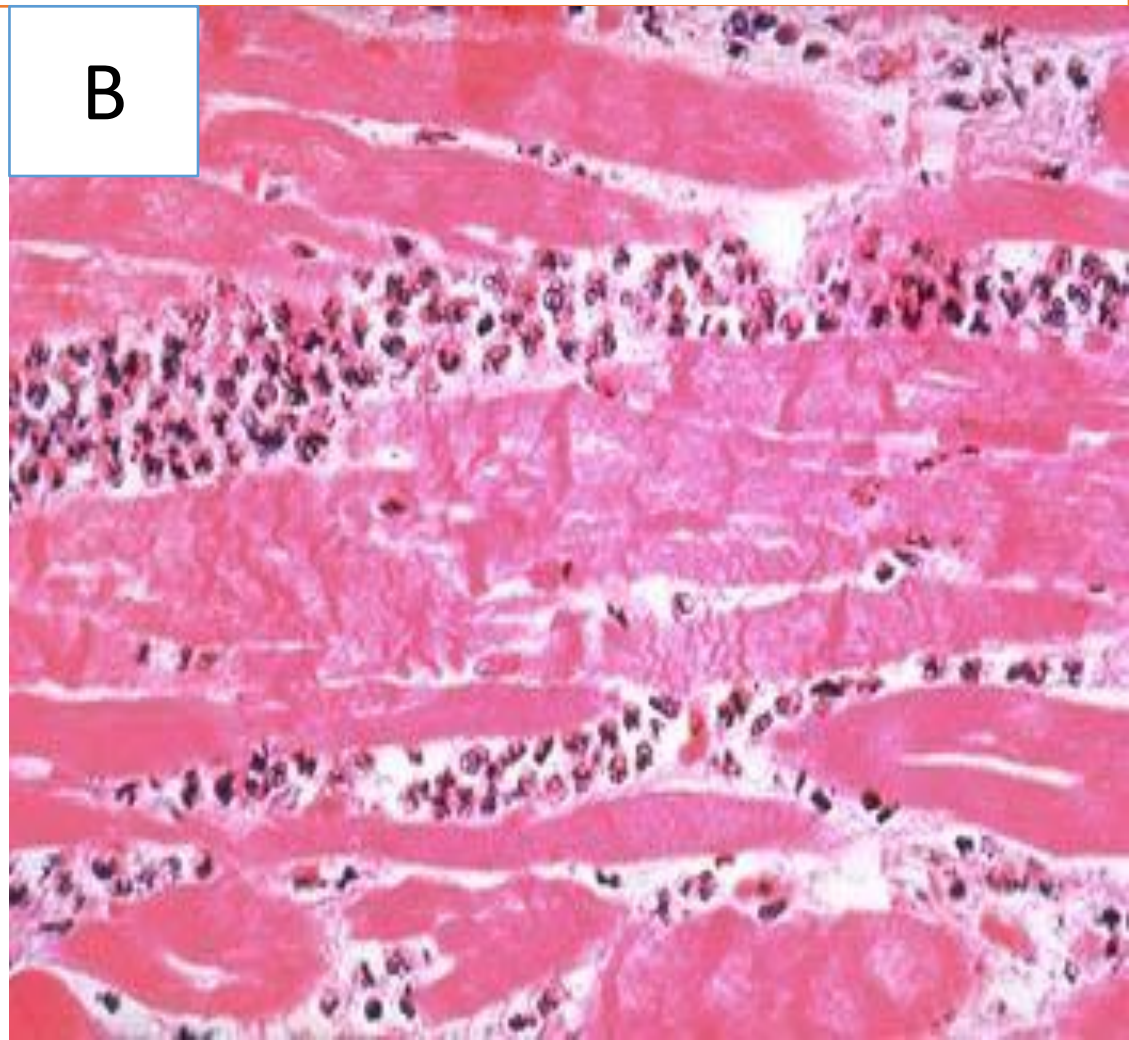
CASE 3

Study a gross (A) and microscopic (B) pictures of heart tissue, taken from a 70-year-old man who died due to heart attack.

A



B



Q1. State the type of necrosis seen in “A” and “B”.

Q2. Describe the underlying mechanism of these pathological change.

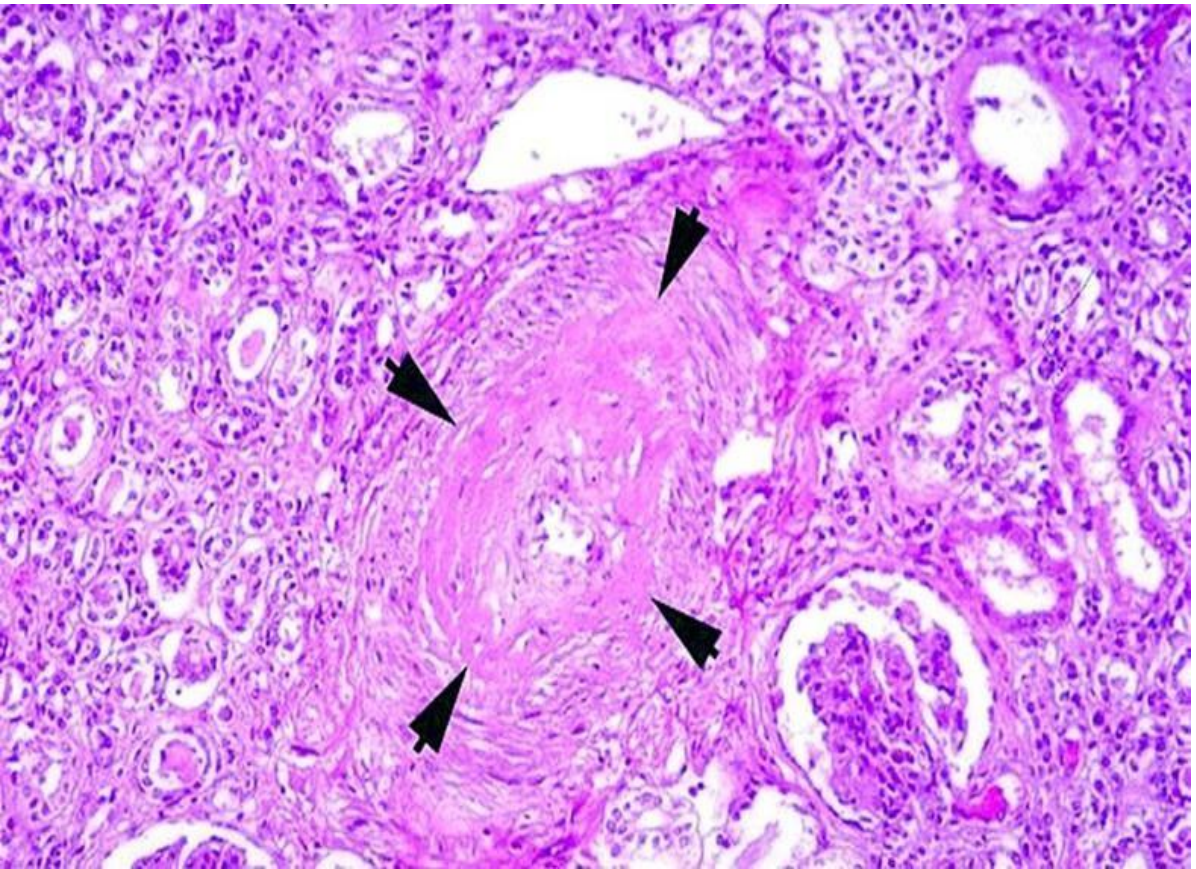
Q3. State TWO OTHER examples of similar pathology.

Q4. Describe the microscopic appearance of necrotic cells in this case.

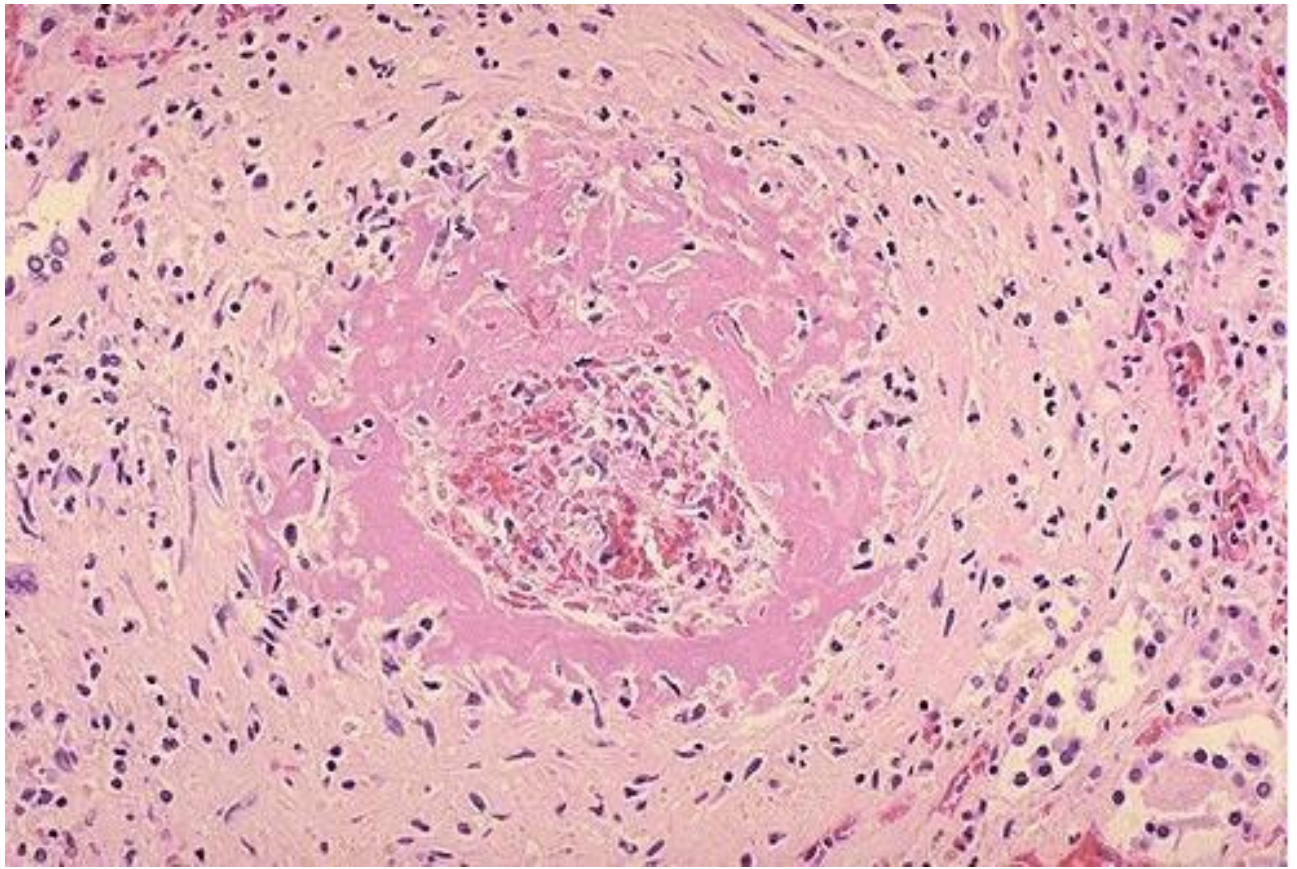
CASE 4

A microscopic examination of renal tissue from a 50-year-old man with malignant hypertension and renal failure.

Low power view



High power view

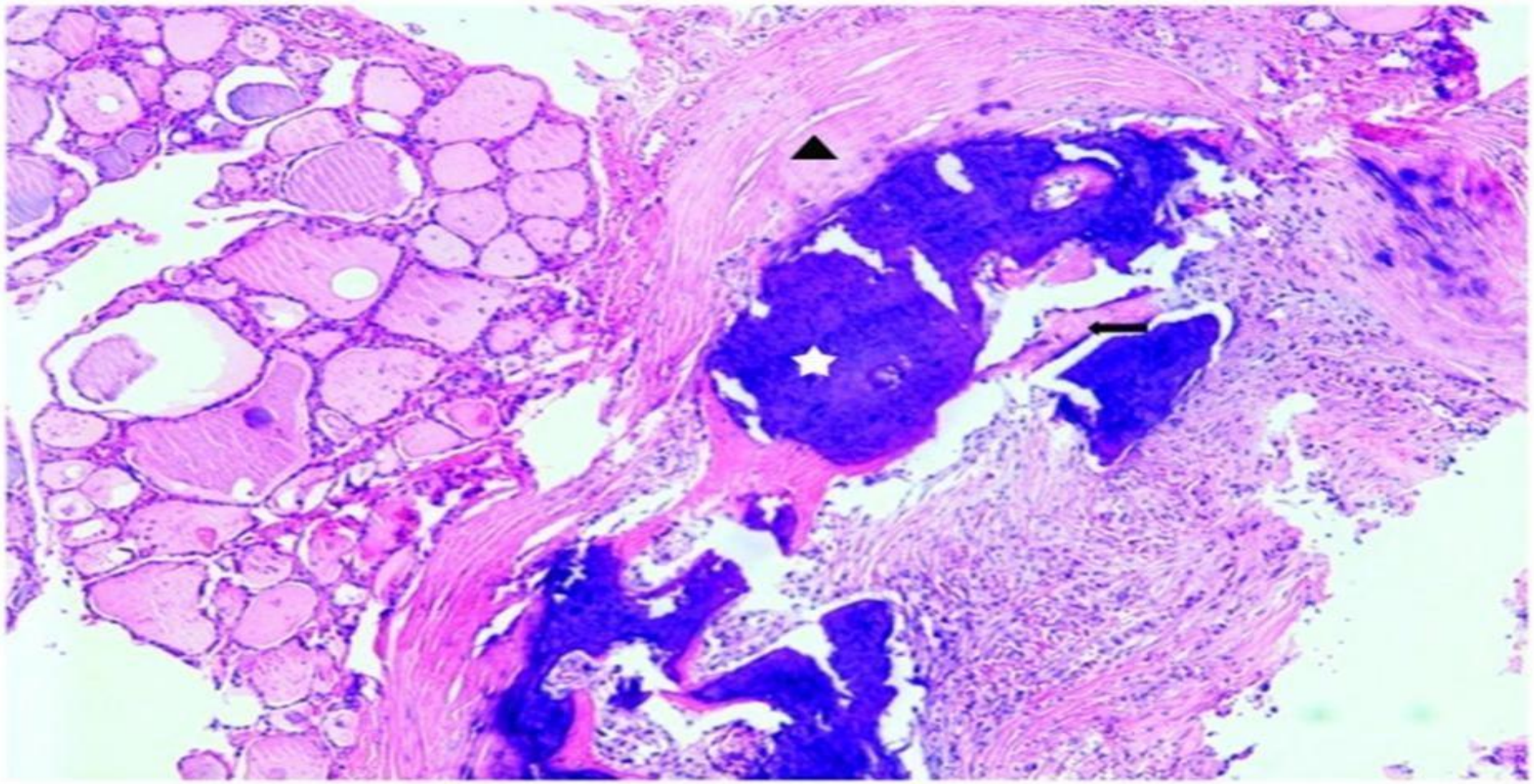


Q1. State the type of necrosis seen in this case.

Q2. Describe the underlying mechanism of these pathological change.

CASE 5

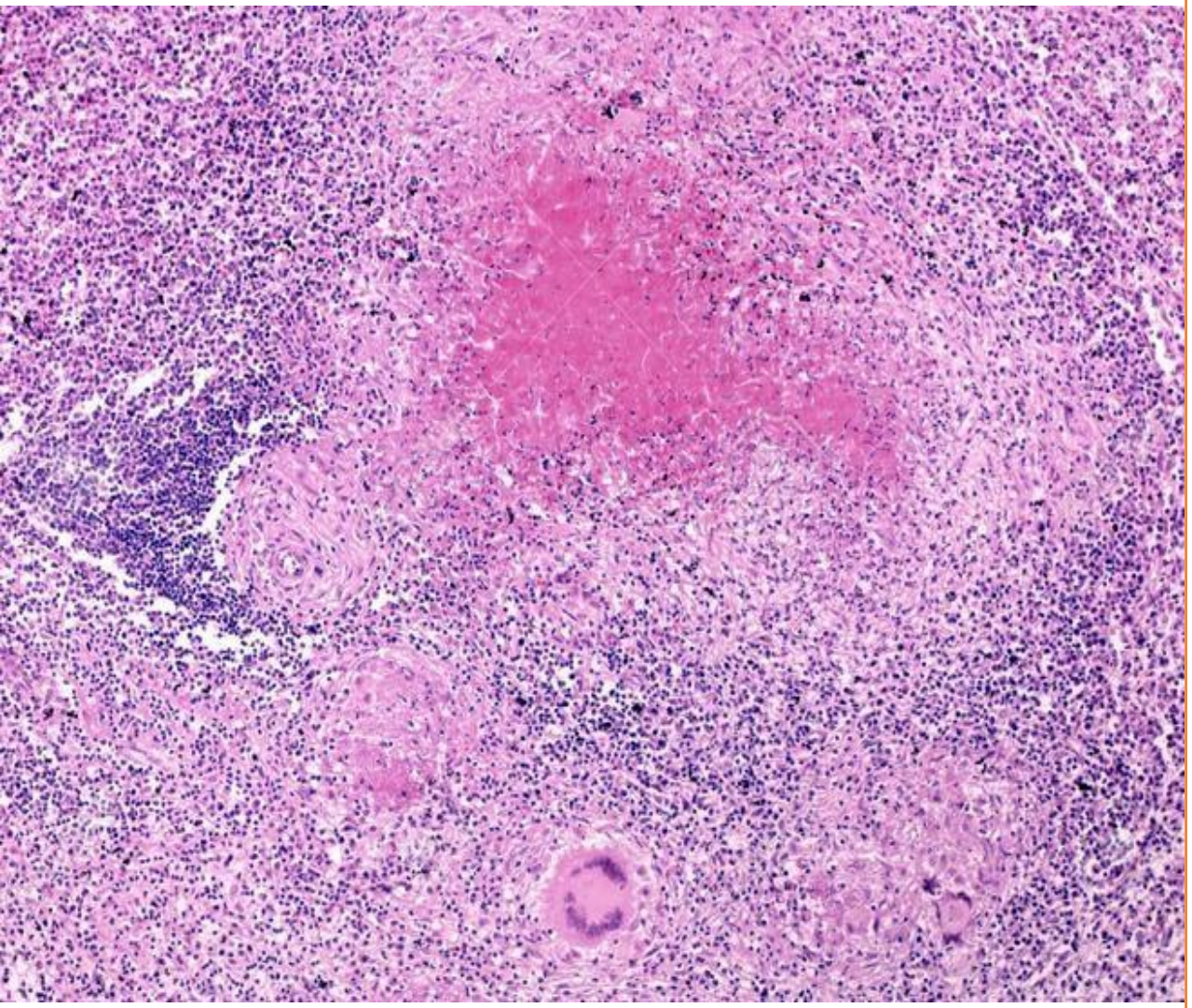
A biopsy of thyroid tissue from a 72-year-old woman with 15 years history of thyroid enlargement.



Q1. State pathological change in this case.

Q2. List the other causes of similar pathological change

CASE 6



Study the microscopic picture of lymph node from a 60-year-old man presented with low-grade fever, weight loss, and persistent cough.
A chest X-ray revealed "consistent with a Gohn complex".

Q1. State the type of necrosis seen in this case.

Q2. Describe the underlying mechanism of these pathological change.

Practical: Cellular adaptation and Tissue degeneration

Case study 1

Describe the gross morphologic features of uterus seen in **A** and **B**.

Gravid uterus



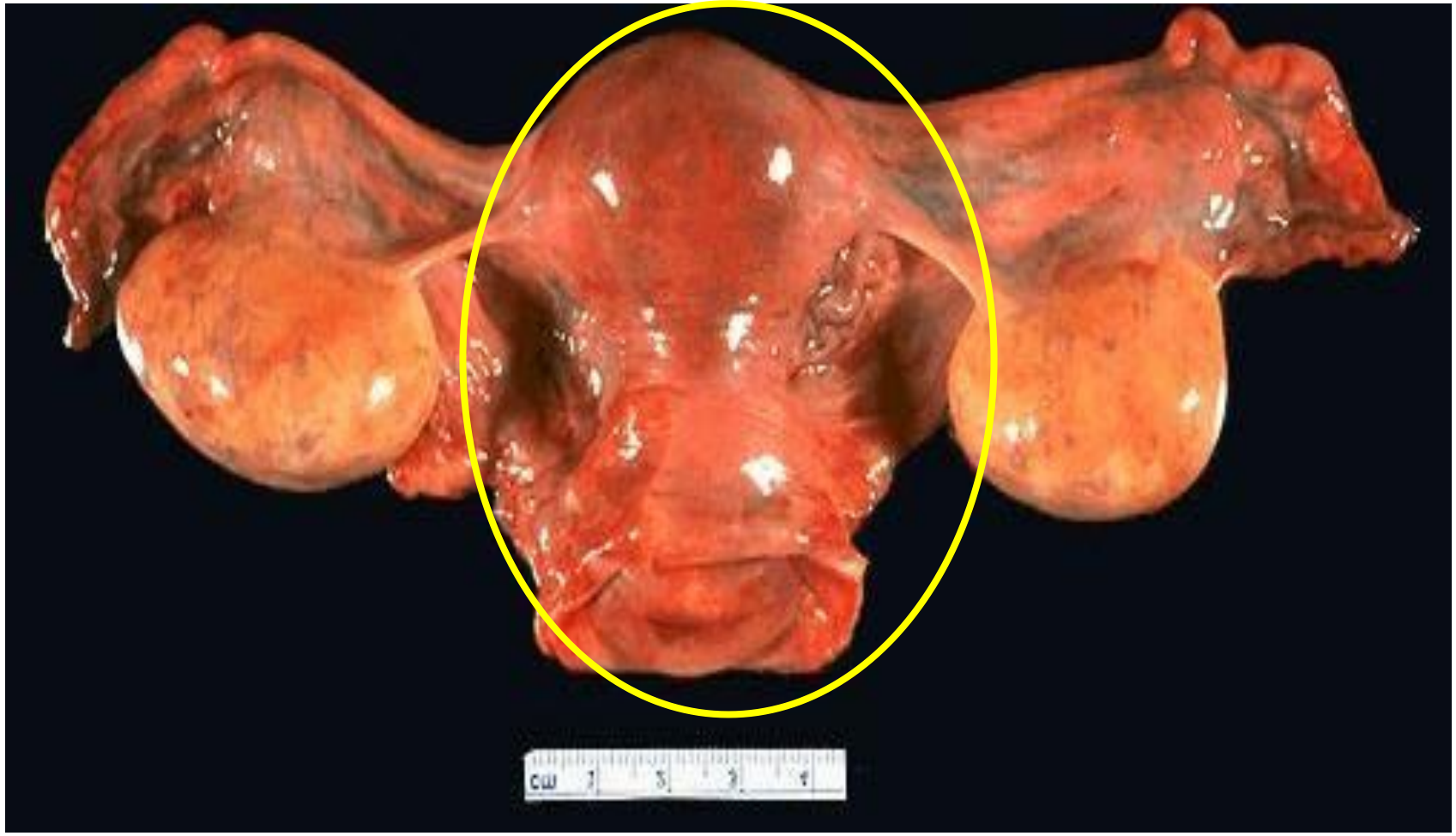
A

A 60-year-old woman uterus



B

Normal Uterus



Gross appearance of a normal uterus with fundus, lower uterine segment, cervix, vaginal cuff, right fallopian tube, left fallopian tube, right ovary, and left ovary from a young woman.

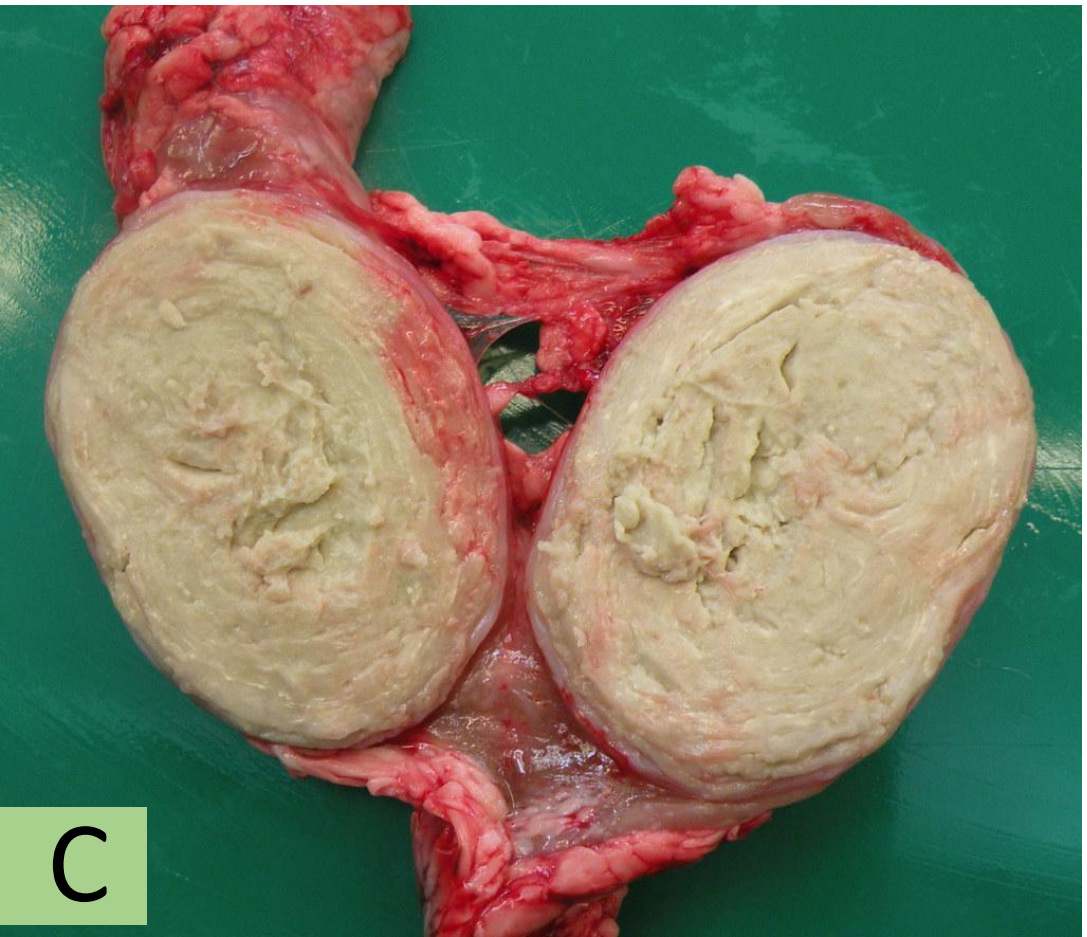
Q1. Describe the gross morphologic features of uterus seen in **A** and **B**.

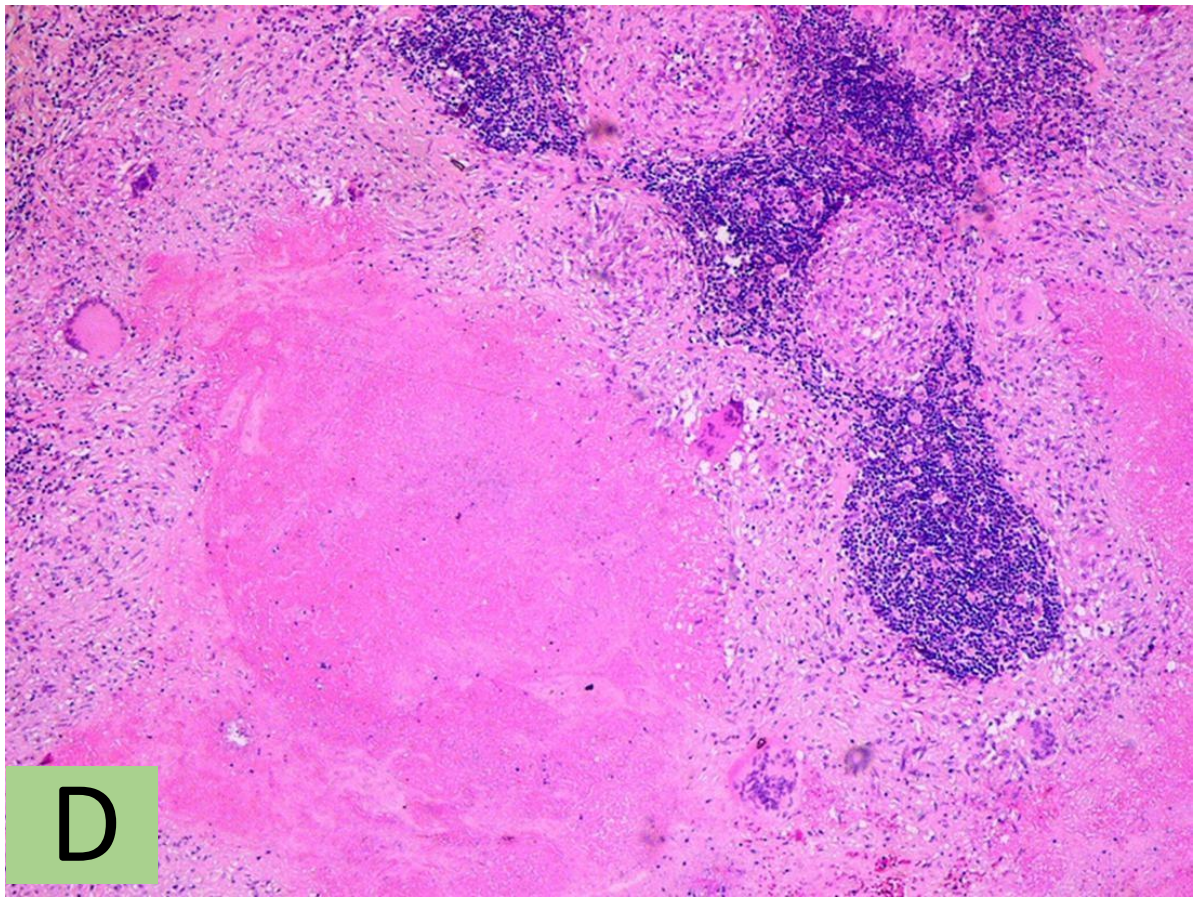
Q2. State the adaptive changes and its cause in **A** and **B**.

Q3: State the molecular mechanisms underlying the cellular adaptations in A and B.

CASE 2

A 30-year old man died due to tuberculosis of the lung with an enlarged lymph node. Study the gross pathology **C** and the microscopic pathology **D**, of his lymph node.





Q1. Describe THREE gross pathology seen in his lymph node C.

Q2. Describe the microscopic pathology of necrosis D, seen in his lymph node.

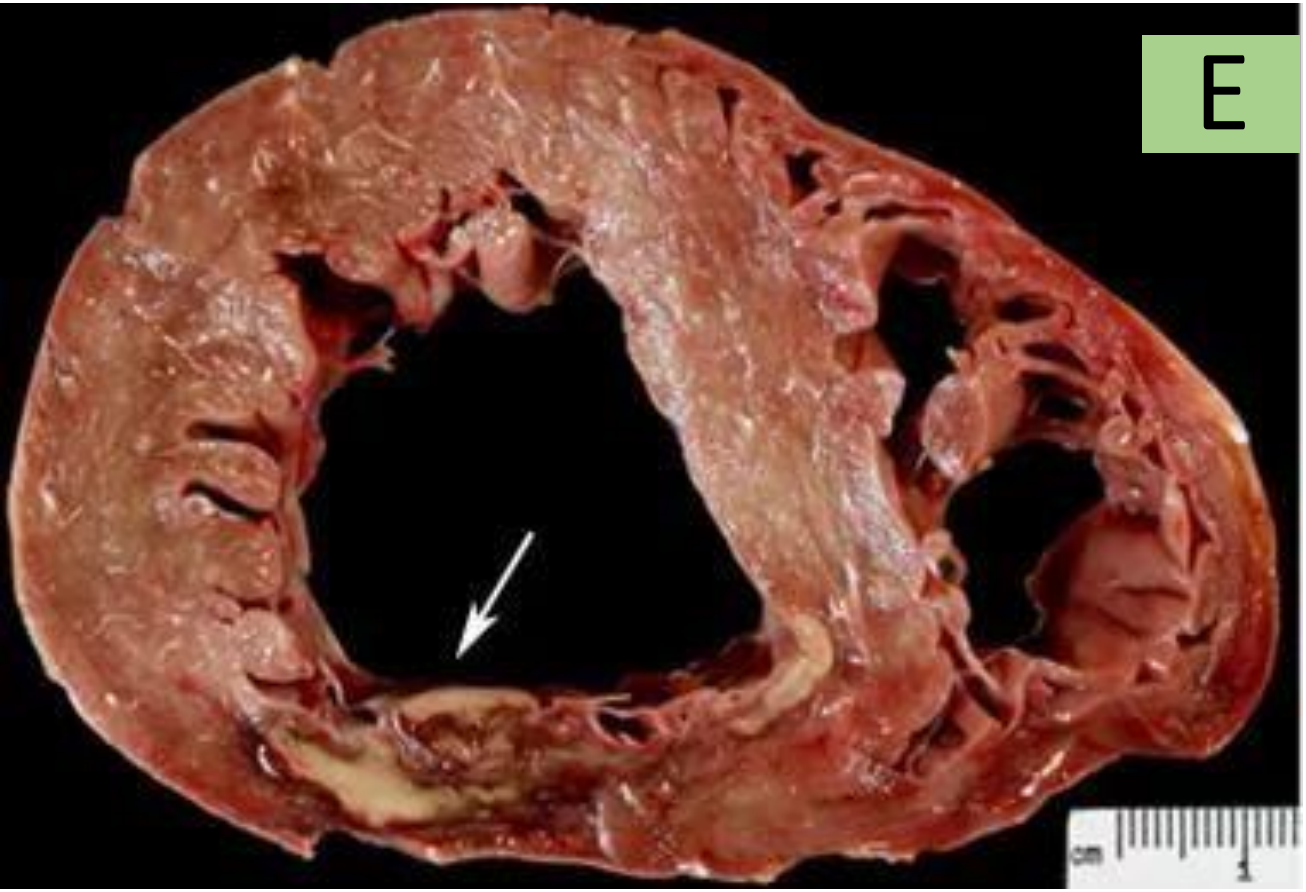
Q3. State the type of necrosis seen.

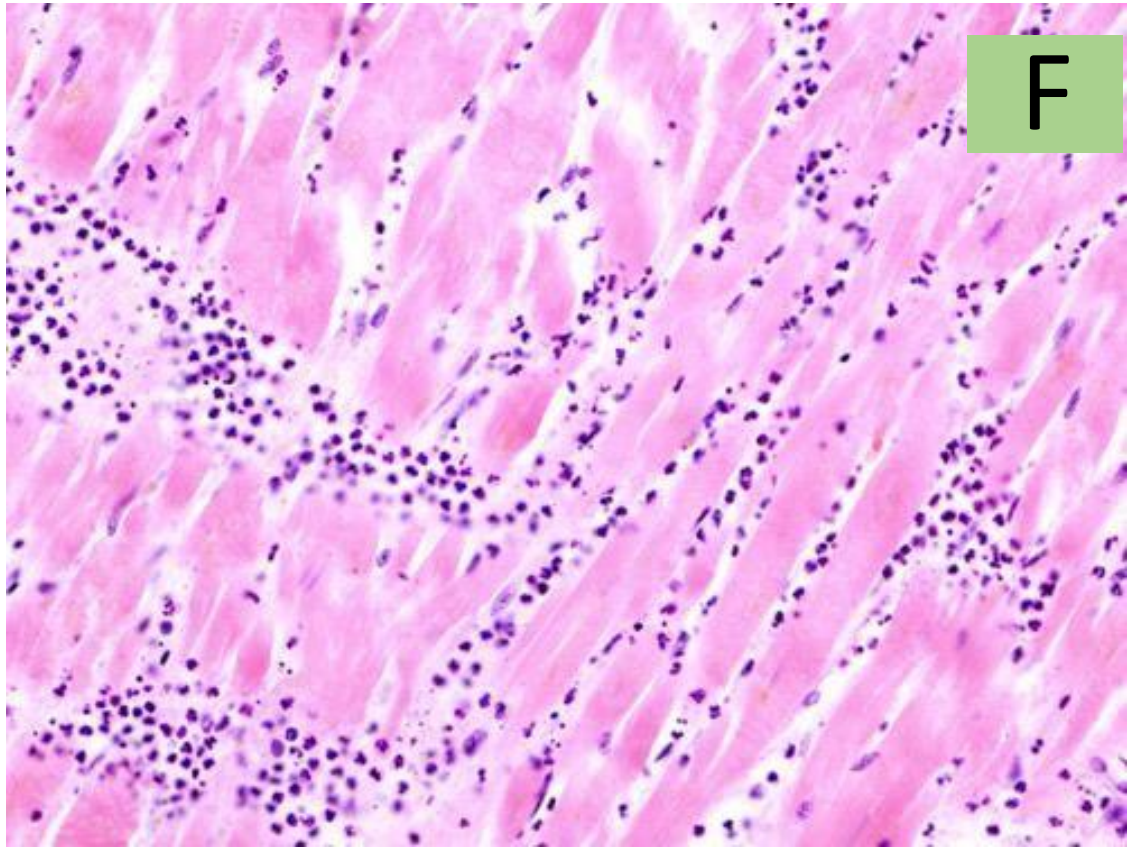
Q4. State one microscopic feature that supports your diagnosis.

CASE 3

A 60-year-old man died from a heart attack (myocardial infarction).

Study the gross pathology **E** and microscopic pathology **F** taken from his heart.





Q1. Describe the gross pathology seen in E.

Q2. Describe the microscopic pathology seen in F.

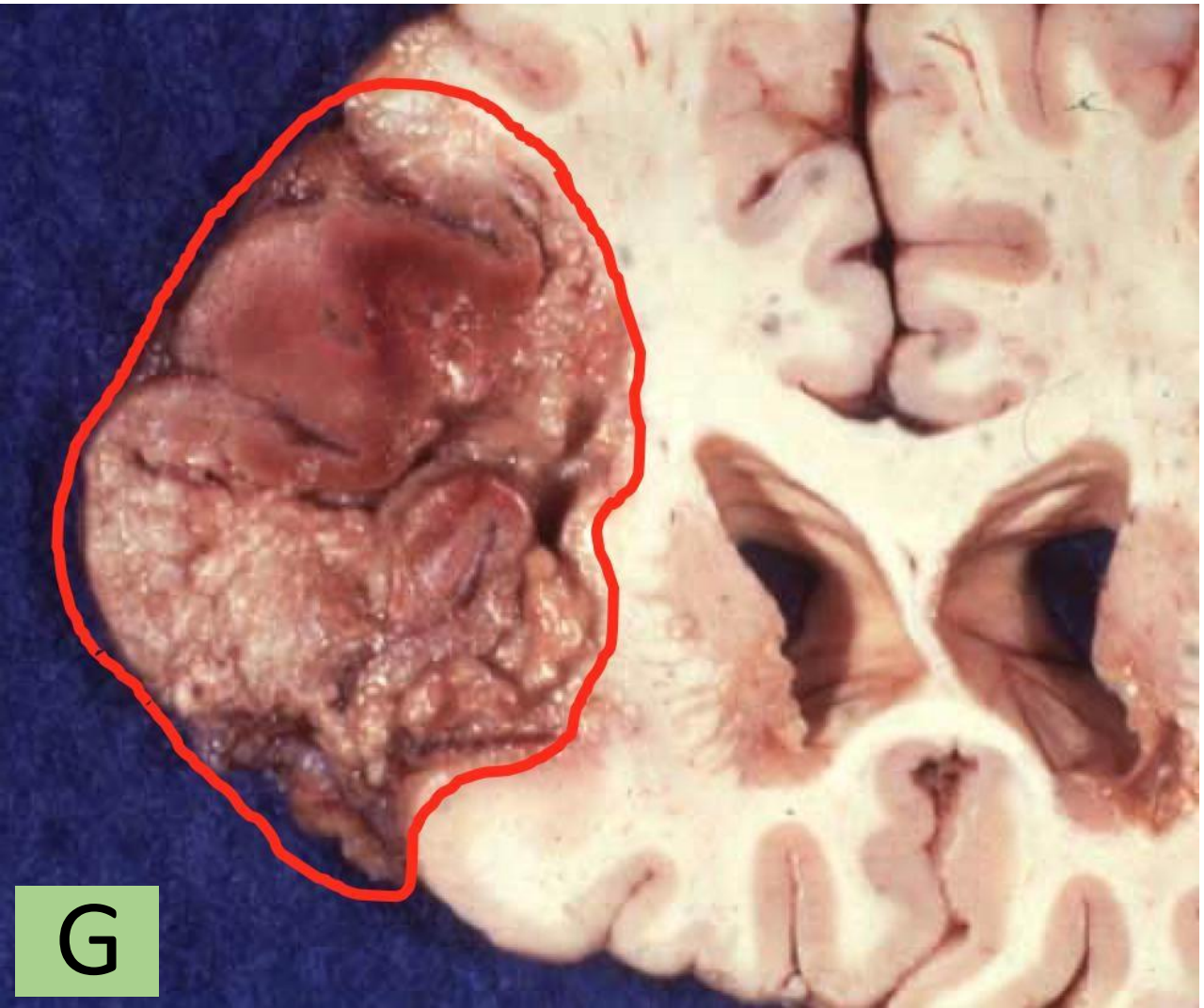
Q3. Draw and label the infarcted myocytes.

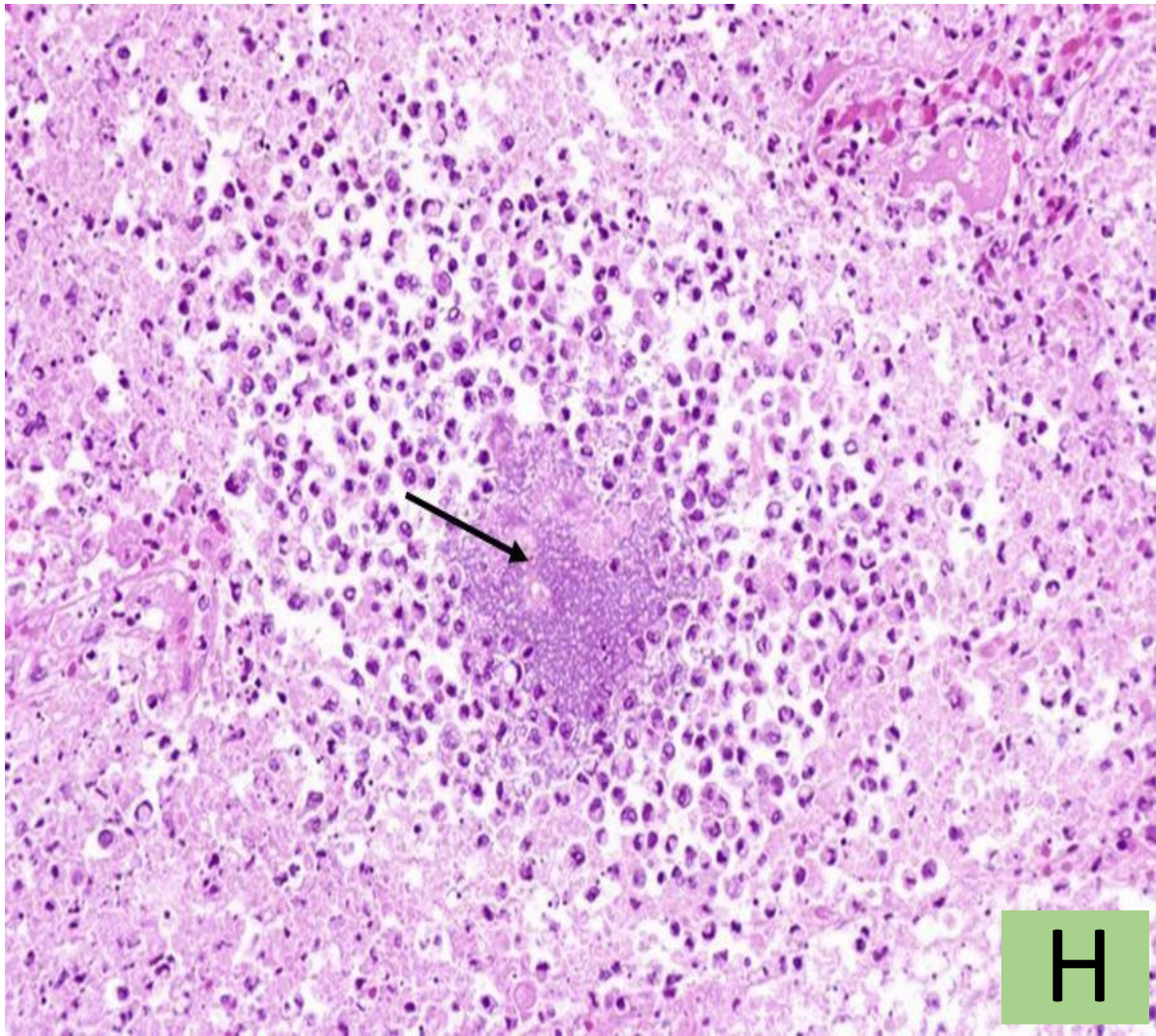
Q4. State the type of necrosis seen.

Q5. State one microscopic feature that supports your diagnosis. Briefly explain the mechanism that produces this microscopic appearance.

Case 4

Study the gross pathology **G**, and microscopic pathology **H** taken from a brain abscess.





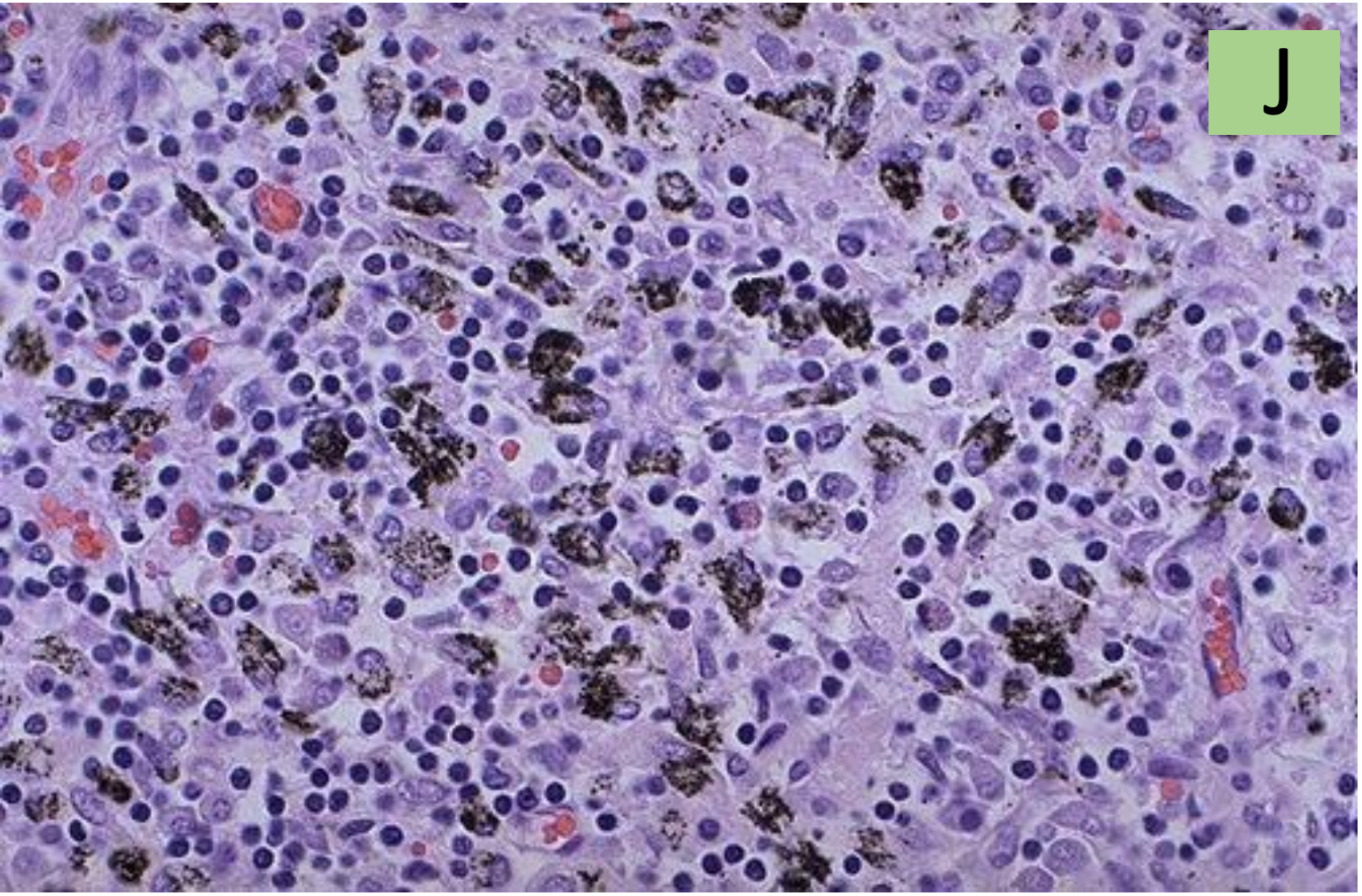
- Q1. Describe the gross pathology seen in **G**.
  Q2: Describe the microscopic pathology seen in **H**.
  Q3. State the type of necrosis seen.
  Q4. State one microscopic feature that supports your diagnosis. Briefly explain the mechanism that produces this microscopic appearance.



Case study 5

Study the lung specimen **I** taken from a city dweller.

Q1. Describe the gross pathology seen in I.



Study the microscopic pathology **J** taken from his lung.

3. Identify the location of the pigments. (intracellular?/extracellular?)

4. Is this an endogenous or exogenous pigment?

5. List other pigments and state whether they are endogenous or exogenous.

Pigments	Endogenous/exogenous

Cell Injury Case Studies and Practical: Answers

Answers based on the MDP 10208 and MDP 10107 lecture notes (Prof Dr Mohammad Zulkarnaen).

Part 1: Cellular Adaptation and Reversible Injury (MDP 10208)

Case 1: Boy with leg in a plaster cast

The question says left leg, but the thinner leg is the right (immobilised) one.

1. **Atrophy**: a decrease in the size of an organ or tissue due to decreased cell size and/or cell number.

2. **Disuse atrophy**, from 6 weeks of immobilisation in the cast. His right leg was immobilized due to fractures of bones and this resulted in disuse of right leg muscles which caused the atrophy of muscles. The initial decrease in cell size is reversible once activity is resumed. With more prolonged disuse, skeletal muscle fibers decrease in number (due to apoptosis) as well as in size.

3. **Mechanism**: protein synthesis falls and protein degradation rises, so the cell's structural components shrink. Degradation goes through two systems: lysosomes (contains acid hydrolase & other enzymes that degrade endocytosed protein from extracellular environment and cell surface) and the ubiquitin-proteasome (UPP) pathway (responsible for degradation of cytosolic & nuclear proteins. The atrophic cell has diminished function but is **not dead**, so it recovers with rehabilitation exercise.

4. **Other causes** (any four): denervation, loss of endocrine stimulation, ageing, pressure (e.g. tumour compression), ischaemia (reduced blood supply), inadequate nutrition.

Case 2: Smoker with haemoptysis

1. **Squamous metaplasia**: a reversible change in which one differentiated adult cell type (ciliated columnar epithelium) is replaced by another (stratified squamous), caused here by chronic irritation from smoking.

2. **Mechanism**: reprogramming of stem cells in the epithelium (or mesenchymal cells in connective tissue), so new cells differentiate along a different pathway. It is fully reversible if the stimulus stops. The cost is loss of function (e.g. ciliary clearance) and an increased risk of malignancy.

Smoking → chronic irritation → loss of mechanisms of protection against infection—mucus secretion and the ciliary action of the columnar epithelium → can alter the normal pseudostratified ciliated columnar epithelium to the squamous epithelium.

3. **Other examples**:
   - **Barrett oesophagus**: squamous to intestinal-type columnar epithelium (in your slides).
   - *(extra)* **Cervix**: columnar to squamous at the transformation zone.
   - *(extra)* **Urinary bladder**: urothelium to squamous with chronic stones or infection (Presence of ova of trematode Schistosoma haematobium)

- *(extra)* **Vitamin A (retinoic acid) deficiency**: squamous change in respiratory epithelium

4. **Two other adaptations**:
   - **Hypertrophy** is an increase in cell size, e.g. cardiac hypertrophy in hypertension, or skeletal muscle in a bodybuilder. Its mechanism is increased protein synthesis and more organelles.
   - **Hyperplasia** is an increase in cell number in mitotic cells, e.g. endometrial hyperplasia from excess oestrogen, or liver regeneration after partial hepatectomy. Growth factors drive mature cells to divide or activate stem cells.

Case 3: High-dose paracetamol, liver biopsy

1. **Cellular changes**: hepatocytes are swollen and pale, with large clear vacuoles in the cytoplasm (**hydropic change**). The mechanism runs like this:
   - Paracetamol is converted to reactive metabolites, which form ROS and cause lipid peroxidation and membrane damage.
   - Mitochondrial injury lowers ATP, so the Na+ pump fails.
   - Na+, water and Ca2+ flow in, and the cell and its ER swell.

2. **Reversible cell injury** (hydropic swelling or degeneration).

3. **Causes of cell injury**: ischaemia/hypoxia, chemical agents and drugs, physical agents, infectious agents, immune reactions, genetic derangements, nutritional imbalances. **Example: acetaminophen overdose**, which causes *indirect* toxicity. Metabolic activation forms reactive metabolites, then ROS, then lipid peroxidation and membrane damage. (Direct toxicity, e.g. mercury, acts straight on membranes and organelles.)

4. **Biochemical mechanisms**:
   - ATP depletion
   - Mitochondrial damage
   - Ca2+ influx and loss of calcium homeostasis
   - Accumulation of oxygen-derived free radicals (oxidative stress)
   - Defects in membrane permeability
   - Damage to DNA and proteins (misfolded proteins)

Case 4: Thalassaemia major, multiple transfiusions

1. **Haemosiderin**, an endogenous pigment. It appears golden-brown on H&E and **blue on Prussian blue** stain, which confirms iron.

2. **Mechanism**: repeated transfusions cause systemic iron overload. Haemoglobin breakdown releases iron, which is stored as ferritin. Aggregates of ferritin micelles form haemosiderin granules in hepatocytes, causing liver enlargement. *(extra)* Excess iron can also generate free radicals and injure the liver, and haemolysis explains the jaundice.

3. **Other pigments**:

- **Lipofuscin** (endogenous): wear-and-tear pigment seen in cardiac myocytes. It is not injurious to the cell.
- **Carbon/coal dust** (exogenous): inhaled, taken up by macrophages, and deposited in lungs and lymph nodes, causing anthracosis.
- **Tattoo pigment** (exogenous) and **melanin** (normal endogenous pigment) are also in your slides.

Part 2: Irreversible Cell Injury (MDP 10107)

Case 1: HIV-positive man, lung abscess

1. **Liquefactive necrosis**.
2. **Three nuclear patterns**:
   - **Pyknosis**: nuclear shrinkage, with DNA condensed into a small dark basophilic mass.
   - **Karyorrhexis**: the pyknotic nucleus's membrane ruptures and the nucleus fragments.
   - **Karyolysis**: nuclear fading, as chromatin dissolves through the action of DNases and RNases.

3. **Mechanism**: bacteria (e.g. *Staphylococcus*, *Streptococcus*, *E. coli*) attract neutrophils. Neutrophils release enzymes to kill the bacteria, and these enzymes also digest the tissue. The tissue softens and liquefies into pus, walled off from healthy tissue as an abscess cavity.

4. **Another example**: cerebral infarct or brain abscess. Brain tissue is rich in hydrolytic enzymes and lipids and has little connective tissue, so cells are digested by their own hydrolases. Wet gangrene is another.

Case 2: Pancreas in acute pancreatitis

1. **Fat necrosis** (enzymatic). (May caused by direct trauma to adipose tissue and extracellular liberation of fat enzymatic lysis of fat due to release of lipases)

2. **Mechanism**: injury to pancreatic acini releases **lipases**. Lipases break triglycerides into fatty acids, which combine with Ca2+, Mg2+ and Na+ to form soaps (**saponification**). This gives the opaque, chalky-white deposits. Under the microscope, cell outlines remain but the fat cells have lost their nuclei and the cytoplasm is pink and amorphous.

• Fat necrosis, result from enzymatic activity of lipase.
• Result from release of activated pancreatic lipases into the substance of the pancreas and peritoneal cavity.
• Pancreatic enzymes leak out of acinar cells and liquefy the membranes of fat cells in the peritoneum.
• The released lipases split the triglyceride esters contained within fat cells.

3. **Another example**: traumatic fat necrosis of the breast. Trauma causes haemorrhage and swelling, then ischaemia and pressure necrosis of fat cells.

Case 3: Myocardial infiarction

1. **Coagulative necrosis**.

2. **Mechanism**: ischaemic hypoxia from obstruction of blood supply (infarction) lowers ATP and

causes acidosis. Acidosis denatures proteins, including enzymes. Denatured enzymes are inactive and cannot digest the cell, so the **tissue architecture is preserved**.

• The injury denatures (d/t increased acidosis) not only structural proteins but also enzymes → blocks the proteolysis of the dead cells.
• As a result of eosinophilic, anucleate dead cells may persist for days or weeks.
• Cause – ischaemic hypoxic injury.

3. **Other examples**: spleen, gut, skeletal muscle (your slide also lists most organs).

4. **Microscopic appearance**: dead myocytes are more eosinophilic, with nuclei lost (karyolysis) or showing pyknosis and karyorrhexis. The cell outlines are preserved and the cytoplasm may be vacuolated. In B there are also neutrophils between the fibres.

Case 4: Kidney in malignant hypertension

1. **Fibrinoid necrosis**: the vessel wall (arteriole) is replaced by bright pink, amorphous material.

2. **Mechanism** (per the slides): immune complexes (antigen-antibody) deposit in the artery wall, activate complement and attract neutrophils. The deposits give the bright pink, amorphous appearance. *(extra)* In malignant hypertension, severe pressure injury to the vessel wall also lets plasma proteins and fibrin leak into it. Follow whichever mechanism your lecturer emphasises.

• Special form of necrosis usually seen in immune reactions blood vessels. This pattern of necrosis typically occurs when complexes of antigens and antibodies are deposited in the walls of arteries.
• Deposits of these "immune complexes," together with fibrin that has leaked out of vessels.
• Result in a bright pink and amorphous appearance in H&E stains, called "fibrinoid" (fibrin-like).

Case 5: Thyroid, 15-year goitre

1. **Dystrophic calcification**. The dark purple, basophilic, amorphous clumps are in damaged tissue. Serum calcium is normal, with no systemic calcium disorder.

Dystrophic calcification in the thyroid tissue.
It is thought that calcium is concentrated in membrane-bound vesicles in cells by a process that is initiated by membrane damage.
Several steps:
(1) Calcium ion binds to the phospholipids present in the vesicle membrane
(2) Phosphatases associated with the membrane generate phosphate groups, which bind to the calcium
(3) The cycle of calcium and phosphate binding is repeated, raising the local concentrations and producing a deposit near the membrane
(4) A structural change occurs in the arrangement of calcium and phosphate groups, generating a microcrystal, which can then propagate and lead to more calcium deposition.
Dystrophic calcification of thyroid results from degenerative changes (calcified colloid and degenerated epithelium, psammoma bodies, old hemorrhage, vessel wall, etc.)

2. **Other causes** (settings where dystrophic calcification occurs):
   - **Dead tissue**: necrosis, infarcts, thrombi, haematoma, dead parasites.
   - **Degenerative tissue**: dense old scars, atheromas, stroma of tumours (e.g. leiomyoma), Monckeberg sclerosis, damaged heart valves.

Case 6: Lymph node, Ghon complex (TB)

1. **Caseous necrosis**.

2. **Mechanism**: the *M. tuberculosis* cell wall contains a complex wax (peptidoglycolipids) that is toxic to cells. Dead cells disintegrate but are **not completely digested** by hydrolases. Grossly this looks cheese-like, soft, grey and friable. Microscopically there is a **granuloma (tubercle) with a central area of caseous necrosis**, ringed by giant cells and lymphocytes.

Combined features of coagulative and liquefactive necrosis
-Denaturation of proteins
-Enzymatic digestion of the cell

Part 3: Practical 2026

Case study 1: Uterus A and B

1. **Gross features**:
   - **A (gravid)**: enlarged, bulky and globular, with a very thick muscular wall and a large cavity containing dark placental or decidual remnants.
   - **B (60-year-old)**: small, shrunken and pale, with a thin wall and a narrow, slit-like cavity.

2. **Adaptive changes**:
   - **A: hypertrophy and hyperplasia** of myometrial smooth muscle (physiological), caused by hormonal stimulation (oestrogen) in pregnancy.
   - **B: atrophy** (physiological), caused by loss of endocrine stimulation and ageing after menopause.

   Cause:
   A: stimulated mainly by increased oestrogen during pregnancy
   B: due to reduced hormonal stimulation, especially reduced oestrogen, after menopause

3. **Molecular mechanisms**:
   - **A, hypertrophy**: increased protein synthesis and increased size and number of organelles. **A, hyperplasia**: growth factors and hormones drive mature cells to divide, or activate tissue stem cells. The uterus can do both because it has dividing cells.
   - **B**: decreased protein synthesis and increased protein degradation via lysosomes and the ubiquitin-proteasome pathway, with loss of cell components. The cells are smaller with reduced function but not dead.

   A: Hormonal stimulation activates growth-promoting genes, resulting in smooth-muscle-cell proliferation (hyperplasia) and increased protein synthesis with increased cell size (hypertrophy).
   B: Oestrogen withdrawal reduces protein synthesis. Increased protein degradation through the ubiquitin–proteasome pathway and increased autophagy reduce cellular proteins and organelles, causing atrophy

Case study 2: TB lymph node (C and D)

1. **Three gross features** of C:
   - The node is enlarged.
   - The cut surface is pale, yellowish-white and cheese-like.
   - It is soft, dry and friable (crumbly) with loss of normal architecture.
   - Thickened fibrous capsule
   - Entirely filled with friable necrotic tissue

- Cheese/yellowish coloured material

2. **Microscopy of necrosis in D**: the centre is a structureless, amorphous, granular pink area with no cell outlines. It is surrounded by granulomatous inflammation: epithelioid macrophages, multinucleated giant cells and clusters of lymphocytes.
3. **Type of necrosis**: **caseous necrosis**.
4. **One supporting feature**: a granuloma (tubercle) with a central area of caseous necrosis and surrounding giant cells.

   Microscopic feature: Central amorphous, granular, eosinophilic, acellular debris with loss of normal tissue architecture, often surrounded by granulomatous inflammation

Case study 3: Myocardial infiarction (E and F)

1. **Gross (E)**: a well-demarcated area of pale yellow-tan, softened myocardium (arrow) in the ventricular wall, sharply separate from the surrounding viable red-brown muscle.

   -Yellow/pale tan necrotic area

   -in the wall of the heart

2. **Microscopy (F)**: dead myocytes are bright pink (hypereosinophilic) with loss of nuclei. There is karyolysis, pyknosis and karyorrhexis in places. Numerous **neutrophils** infiltrate between the fibres, and the fibre outlines are preserved.

3. **Drawing to make**: draw 2 or 3 parallel myocyte fibres and label:
   - necrotic myocyte, with pink cytoplasm and lost striations
   - karyolysis (absent or faded nucleus)
   - pyknotic nucleus
   - karyorrhexis (fragments)
   - neutrophils in the interstitium
   - one normal myocyte with a central oval nucleus for comparison

4. **Type of necrosis**: **coagulative necrosis**.

5. **Supporting feature**: hypereosinophilic myocytes with preserved outlines and absent nuclei. Ischaemia lowers ATP, and anaerobic glycolysis builds up lactate and lowers pH. Acidosis denatures proteins, which turns the cytoplasm pink and coagulated, like cooking egg white. Denatured enzymes cannot digest the cell, which is why the outlines remain. Nuclei are lost through DNases and RNases (karyolysis).

   Preservation of cell/tissue outlines despite loss of nuclei

   Explain: Ischaemic injury causes denaturation of structural proteins and lysosomal enzymes. Enzyme proteolysis is therefore inhibited, so the outlines of dead cells and tissue architecture are preserved temporarily

Case study 4: Brain abscess (G and H)

1. **Gross (G)**: a circumscribed lesion (outlined) where the brain tissue is destroyed and replaced by soft, yellow-tan purulent and necrotic material with red-brown haemorrhagic areas. The normal gyral pattern is lost and the lesion has a mass effect.

2. **Microscopy (H)**: sheets of neutrophils among pink, amorphous, granular necrotic debris, with no preserved tissue architecture. The arrow marks a dense purple clump, likely a bacterial colony.

3. **Type of necrosis**: **liquefactive necrosis**.

4. **Supporting feature**: sheets of neutrophils (pus) in structureless debris. Neutrophils release hydrolytic enzymes to kill bacteria, and these enzymes also digest the tissue. The brain is rich in hydrolases and lipids and has little connective tissue, so cells are digested by their own enzymes, and the tissue softens and liquefies.

   Microscopic feature: Loss of normal cerebral tissue architecture, replaced by abundant neutrophils and granular cellular/necrotic debris around the fungal ball

   Mechanism: Hydrolytic enzymes released from neutrophils and lysosomes digest dead cells and extracellular tissue. This enzymatic digestion liquefies the brain tissue, producing liquefactive necrosis

Case study 5: City dweller's lung (I and J)

1. **Gross (I)**: numerous black, speckled and patchy pigment deposits scattered throughout the lung parenchyma and around the airways, with the rest of the lung red-brown. This is **anthracosis**.

   -Multiple small black macules (holes)/areas of pigment

   -in the lobules of the lungs

2. **Microscopy (J)**: macrophages with abundant black granular pigment in their cytoplasm, mixed with lymphocytes. There is no necrosis.

3. **Location**: **intracellular**, within macrophages.

4. **Type**: **exogenous** (inhaled carbon).

5. **Other pigments**:

Pigment	Endogenous / Exogenous
Lipofuscin	Endogenous
Iron form haemosiderin	Endogenous
Melanin	Endogenous
Ferritin form hemosiderin	Endogenous

Carbon (coal dust)	Exogenous
Tattoo pigment	Exogenous